ONLINE RETAIL PERFORMANCE REPORT

RetailHub UK — Sales, Customer & Return Intelligence Study

End-to-End Analysis: Power BI · Power Query · DAX

PREPARED BY

Emmanuella Danso | Data Analyst

CONTACT

Email: nuelladee7@gmail.com

LinkedIn: www.linkedin.com/in/emmanuella-danso-699b54404

GitHub: https://github.com/emmanuella-danso

DATASET

Online Retail Dataset (UCI Machine Learning Repository) — adapted as RetailHub UK business simulation

DATE

April 2026

£10M	-£894K	9.17%	23.86%	9.95%
Total Revenue	Total Returns	Return Rate	Top 5 Customer Share	Top 10 Product Share

1. EXECUTIVE SUMMARY

RetailHub UK operates in a highly competitive online retail environment where customer behaviour, product performance, and operational efficiency directly influence profitability. This analysis evaluates sales performance, customer value distribution, product contribution, and return behaviour across a **£10M revenue dataset** to uncover actionable insights that support strategic decision-making.

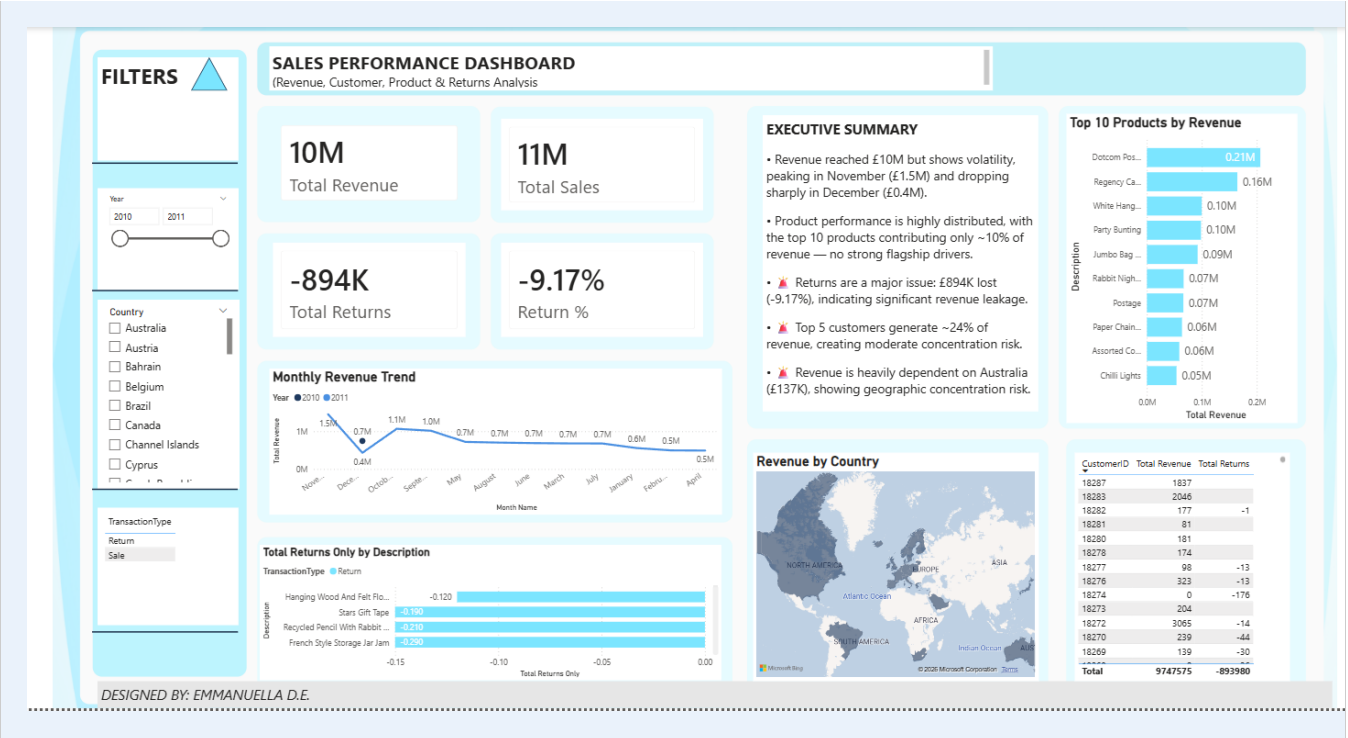
Three critical business challenges were identified that require immediate attention:

01	**Return-Driven Revenue Erosion —** £894K in returns at a 9.17% return rate, significantly above the 5–7% industry benchmark.
02	**Customer Concentration Risk —** top 5 customers contribute 23.86% of total revenue, creating dangerous dependency on a small customer base.
03	**Product Portfolio Imbalance —** top 10 products contribute only 9.95% of revenue, indicating a long-tail distribution with high product dependency risk.

This report delivers a consulting-grade evaluation of RetailHub UK's performance using Power BI, with clear strategic recommendations to strengthen revenue stability, reduce returns, and improve customer retention.

Dashboards

Sales Performance Overview Dashboard



2. BUSINESS OBJECTIVE

RetailHub UK engaged this analysis to answer five critical business questions:

- How much revenue is being generated, and how much is being lost through returns?
- Which customers and products contribute most to revenue — and which represent concentration risk?
- What patterns exist in return behaviour, and what is their quantified financial impact?
- Where are the operational inefficiencies driving above-benchmark return rates?
- What specific, data-backed actions can RetailHub UK take in the next 90 days to improve profitability?

3. DATA PREPARATION & METHODOLOGY

This project was completed entirely in **Power BI** without external modelling tools. All data cleaning, transformations, and calculations were performed using **Power Query** and **DAX**.

3.1 Data Cleaning (Power Query)

- Removed duplicate invoice records to prevent revenue double-counting
- Standardised date formats for accurate time-series analysis
- Corrected negative quantities to properly flag and isolate return transactions
- Ensured consistent product and customer identifiers across all records

- Filtered invalid or missing invoice numbers that would distort KPI calculations

3.2 Feature Engineering

- **Returns Flag** — binary field separating sales transactions from return transactions
- **Revenue** — gross sales value before returns
- **Net Revenue** — revenue after deducting return amounts
- **Return Amount** — total value of returned transactions
- **Customer Contribution %** — each customer's share of total revenue
- **Product Contribution %** — each product's share of total revenue

3.3 DAX Measures Used

DAX Measure	Purpose
Total Revenue	SUM of all positive transaction values
Total Returns	SUM of all negative (return) transaction values
Return Rate	Total Returns / Total Revenue x 100
Net Revenue	Total Revenue + Total Returns
Top 5 Customer Share	Revenue from top 5 customers / Total Revenue
Top 10 Product Share	Revenue from top 10 products / Total Revenue

3.4 Power BI Modelling Approach

- Cleaned tables connected through Power BI's relationship model
- Star schema structure with transactions as the central fact table
- DAX measures built for KPIs, contribution analysis, and return metrics
- Visuals designed for executive-level interpretation with drill-through capability

4. KEY PERFORMANCE INDICATORS

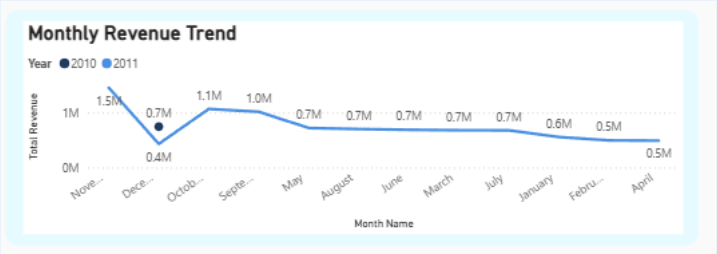
The following KPIs were calculated using DAX and form the foundation of the analysis:

KPI	Value
Total Revenue	**£10,000,000**
Total Returns	**-£894,000**
Return Rate	**9.17%**
Net Revenue	**£9,106,000**
Top 5 Customers Revenue Share	**23.86%**
Top 10 Products Revenue Share	**9.95%**

5. ANALYSIS & KEY FINDINGS

5.1 Revenue Performance

Revenue Trend Dashboard Visual



BUSINESS INSIGHT

RetailHub UK generated **£10M in total revenue** across the analysis period, demonstrating strong sales volume. However, **£894K in return-driven revenue erosion** reduces net revenue to **£9.1M** — an 8.94% reduction in effective revenue before any operating costs are considered. A **9.17% return rate** sits significantly above the retail industry benchmark of 5–7%, signalling either product quality issues, inaccurate product descriptions, or fulfilment errors at scale.

Reducing the return rate by just **2 to 3 percentage points** would recover an estimated **£200K to £300K** in annual revenue — without acquiring a single new customer or launching a single new product.

5.2 Customer Concentration Analysis

Customer Tier	Revenue Contribution
Top 5 Customers	**23.86% of total revenue**
Top 10 Customers	**Estimated 35–40% of total revenue**
Remaining Customers	**Majority of base, minority of revenue**

BUSINESS INSIGHT

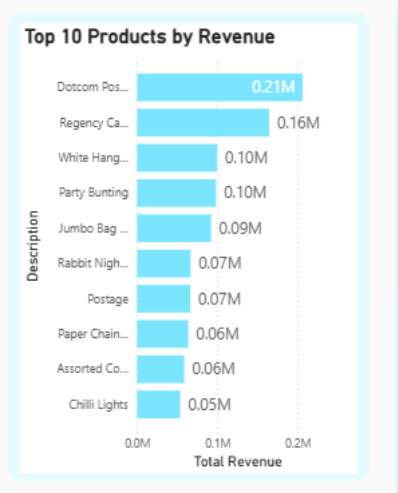
The concentration of **23.86% of total revenue in just 5 customers** creates severe business fragility. Losing even one top customer would have an immediate and material impact on RetailHub UK's revenue. This is not a healthy distribution — it is a concentration risk that should be treated as a board-level strategic concern.

At £10M total revenue, the top 5 customers collectively contribute approximately **£2.39M**. RetailHub UK must implement a structured **key account management programme** to

protect these relationships while simultaneously developing its mid-tier customer base to reduce dependency.

5.3 Product Contribution Analysis

Product Revenue Contribution Chart



BUSINESS INSIGHT

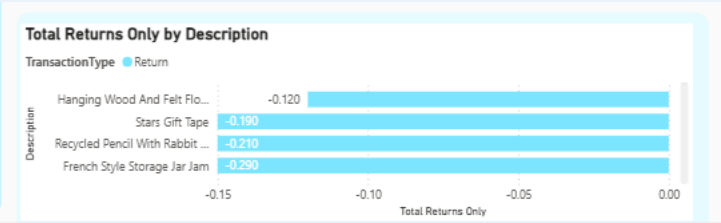
The top 10 products contribute only **9.95% of total revenue**, indicating a highly distributed **long-tail product portfolio**. While this reduces single-product dependency, it creates inventory management complexity and marketing dilution — resources are spread thinly across a large product catalogue rather than concentrated on proven high performers.

Understanding which specific products drive the highest return rates is critical. If the top returned products overlap with the top revenue products, RetailHub UK faces a compounding problem — its best-selling items may also be its most damaging to net revenue. A product-level return rate analysis within Power BI would immediately identify this overlap.

5.4 Return Behaviour & Financial Impact

£894K	9.17%	5–7%	£300K
Total Returns Value	Return Rate	Industry Benchmark	Max Recoverable Revenue

Returns Analysis Dashboard Visual



BUSINESS INSIGHT

A 9.17% return rate represents **a 2.17 to 4.17 percentage point premium** above the industry benchmark range of 5–7%. This excess above benchmark translates to an estimated **£217K to £417K in avoidable return losses** — money that should be generating margin but is instead being consumed by return processing, restocking, and fulfilment costs.

The root cause of above-benchmark returns typically falls into three categories: **product quality failures**, **inaccurate product descriptions or images**, and **fulfilment errors**. RetailHub UK must identify which category is driving its excess returns before investing in corrective action — the solution differs significantly depending on the root cause.

6. KEY BUSINESS RISKS

!	**Critical**	Return rate of 9.17% is 2.17 to 4.17 points above industry benchmark, costing an estimated £217K to £417K in avoidable annual revenue loss.
!	**Critical**	Top 5 customers represent 23.86% of revenue (£2.39M). Loss of even one creates an immediate material revenue gap with no short-term replacement.
!	**High**	Long-tail product portfolio creates inventory complexity. Without product-level profitability analysis, low-margin SKUs may be consuming resources while destroying value.
!	**High**	No systematic return reason tracking. Without knowing why customers return products, all corrective action is guesswork with no measurable ROI.
!	**Medium**	Mid-tier customer base underdeveloped. Revenue over-reliance on top customers leaves RetailHub UK vulnerable to any shift in key account purchasing behaviour.

7. STRATEGIC RECOMMENDATIONS

1

Reduce Return Rate to Industry Benchmark

A 9.17% return rate is costing RetailHub UK an estimated **£217K to £417K per year** above what would be lost at the industry benchmark. The immediate priority is

identifying the **top 10 most-returned products by volume** and auditing their product descriptions, imagery, and sizing information for accuracy.

Implement a **mandatory return reason field** at the point of return — even three categories (wrong product, quality issue, changed mind) would provide enough data to prioritise corrective action. Reducing the return rate from 9.17% to 7% would recover approximately **£217K in annual net revenue** without any new customer acquisition spend.

2 Protect and Develop High-Value Customer Relationships

The top 5 customers generating £2.39M require a dedicated **key account management programme** — dedicated account managers, preferential terms, early product access, and quarterly business reviews. These relationships are too valuable to manage reactively.

Simultaneously, RetailHub UK must identify the **next tier of 10 to 20 high-potential customers** and actively develop them toward higher purchase frequency and basket value. This reduces concentration risk while growing total revenue. A Power BI cohort analysis tracking customer purchase frequency trends over 6 months would identify this development tier immediately.

3 Conduct Product Portfolio Profitability Review

With the top 10 products contributing only 9.95% of revenue, RetailHub UK is operating with **extreme product fragmentation**. A full product-level profitability analysis — revenue minus return costs minus fulfilment costs — would immediately identify which SKUs are generating genuine margin versus which are generating revenue while destroying profit.

Products with high return rates relative to their revenue contribution should be reviewed for discontinuation or description improvement. **Concentrating marketing spend on the top 20 to 30 products by net profitability** — rather than gross revenue — would significantly improve overall portfolio efficiency.

4 Implement Systematic Return Tracking in Power BI

RetailHub UK currently has no structured return reason tracking, meaning all corrective action is based on assumption rather than evidence. Building a **Returns Intelligence Dashboard in Power BI** — tracking return rate by product, category, customer, and time period — would transform returns from an unmanaged cost centre into a managed, improvable metric.

Key metrics to track monthly: overall return rate vs 9.17% baseline, return rate by product category, top 10 returned products by volume and value, return rate by customer segment, and trend vs prior period. A consistent 0.5 percentage point

reduction per quarter would recover **£45K in additional net revenue annually** from each incremental improvement.

8. CONCLUSION

RetailHub UK demonstrates **strong revenue generation capability at £10M**, but is currently constrained by two structural weaknesses: a **9.17% return rate** eroding nearly £900K in revenue annually, and dangerous **customer concentration** with 23.86% of revenue dependent on just 5 customers. These are not minor optimisation challenges — they represent the primary barriers to transitioning from a volume-driven to a sustainably profitable business model.

The data-backed path forward is clear: **reducing the return rate to the 7% industry benchmark** recovers £217K annually. **Protecting top 5 customer relationships** secures £2.39M in at-risk revenue. **Conducting a product portfolio profitability review** eliminates margin-destroying SKUs. Together, these three interventions represent a realistic, achievable path to significantly improved profitability — without requiring a single additional pound of customer acquisition spend.

All DAX measures and Power Query transformation steps are documented within the Power BI file available in the project GitHub repository.

GitHub: https://github.com/emmanuella-danso